

AI in the Workplace: Rapid Review of Skill Transformation in the Industry

1. Introduction

Integrating Artificial Intelligence (AI) in the workplace has led to significant transformations across various industries. This review aims to explore the impacts of AI on the job market, specifically focusing on the skills required to implement and manage AI systems and the human-machine interactions necessary for the current and future workforce.

The importance of this study lies in addressing the skills gap that arises as industries rapidly adopt AI technologies. Understanding the specific skills needed and identifying the challenges faced by businesses and employees can help develop strategies to bridge this gap, thereby ensuring a smooth transition into an AI-augmented work environment.

Objectives:

- To examine how businesses are integrating AI technologies into their workforce.
- To identify the skill sets required for effective AI adoption.
- To analyze the skill gaps or challenges faced in adopting AI.
- To explore strategies employed to address these challenges.

2. Research Questions

- RQ1: How are businesses integrating AI technologies into their workforce?
- RQ2: What are the skill sets required for AI adoption?
- RQ3: What are the skill gaps or challenges faced in adopting AI?
- RQ4: What are the strategies employed to address these challenges?

3. Search Strategy

The search for relevant literature will be conducted using the Scopus database, which has been selected for its comprehensive coverage of scientific publications. The search will be limited to conference papers and journal articles published from 2020 onwards and written in English. This time frame ensures the review captures the most recent developments in integrating AI in the workplace.

The search terms will be formulated to target studies that discuss AI technologies, their implementation in various industries, and the impact on required skills and workforce transformation. Specifically, the search will include terms such as *Artificial Intelligence*, *workforce*, *skills*, *gap*, *challenges*, *strategies*, *solutions*, and *obstacles*. Table 1 presents the final string applied in Scopus on Nov. 14, 2023.

Table 1. Search String for Scopus database.

TITLE ((ai OR “Artificial Intelligence” OR “Machine Learning” OR robot* OR automation) AND (workforc* OR employ* OR skill*) AND (gap* OR challeng* OR obstacl* OR strateg* OR soluti* OR future*)) AND PUBYEAR > 2019 AND (LIMIT-TO (LANGUAGE , “English”)) AND (LIMIT-TO (DOCTYPE , “ar”) OR LIMIT-TO (DOCTYPE , “cp”))
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Inclusion Criteria:

- Articles published from 2020 or newer.
- Written in English.
- Focus on AI integration in various industries.
- Discuss skills required, challenges faced, and strategies for AI adoption.

Exclusion Criteria:

- Articles not focused on AI in the workplace.
- Publications prior to 2020.
- Non-English articles.
- Articles that do not discuss skills, challenges, or strategies.

4. Selection Procedure

The selection procedure will begin with an initial search using the predefined search terms in the Scopus database.

In the first screening phase, the titles and abstracts of the retrieved articles will be reviewed independently by two reviewers to exclude studies irrelevant to the research questions. Studies that meet the inclusion criteria will proceed to the full-text screening phase. During the full-text screening, a single reviewer will assess the full texts of the potentially relevant articles.

A PRISMA flow diagram will be utilized to document the selection process. This diagram will detail the number of articles identified through the database search, the number of articles screened, the number assessed for eligibility, and the number included in the final review.

5. Data Extraction

The data extraction will use a predefined form to ensure consistency and comprehensiveness in capturing relevant information from each selected study. Table 2 presents the data extraction form.

Table 2. Extraction Form.

Field	Description
Author(s)	Records the author(s) who published the study.
Year of publication	Records the publication year of the study.
Title	Records the study's title.
Journal/conference name	Records the journal or conference where the study was published.
Type of Study	Describe the type of study (empirical, theoretical, etc.)
Industry/sector focus	Describe which industry or sector the study focuses on.
AI Integration Details	Describes the types of AI technologies implemented and the scope of their integration within the organization.
Skill Sets Required	Identifies the specific technical and non-technical skills necessary for AI adoption, as discussed in the study.
Challenges Faced	Summarizes the skill gaps and other challenges encountered when adopting AI technologies.
Strategies Employed	Details organizations' approaches and strategies to address the identified challenges and skill gaps.
Outcomes and Impacts	Explores the effects of AI integration on the workforce, including impacts on productivity and organizational performance.

6. Data Synthesis

The data synthesis will focus on qualitative analysis, given that the study primarily aims to understand the thematic and contextual aspects of AI integration in the workplace. The qualitative synthesis will involve a thematic analysis to identify and interpret patterns related to AI integration, required skills, challenges, and strategies across the included studies. The process will include the following steps:

Theme Development: The initial codes will be reviewed and grouped into broader themes. These themes will reflect recurring patterns and critical concepts identified across the studies.

Theme Analysis: Each theme will be analyzed in depth to understand its implications and how it relates to the research questions. The analysis will consider the context provided by the original studies and any variations in findings across different industries or regions.

7. Results

Figure 1 shows the selection procedure results, according to the PRISMA flow diagram. The search was made in Scopus on November 14, returning 39 documents. We excluded one article due to lack of access. The remaining 38 documents were screened by a solo researcher, analyzing titles – excluding five documents – and abstracts – excluding four documents. The remaining articles were analyzed for eligibility and focused on their alignment with the research questions. We excluded 9 papers for not answering the research questions: 7 failed to discuss AI in the workplace, and another 2, although included in the discussion of AI in the workplace, failed to present discussions about skills, challenges, or strategies for AI adoption. At the end of the analysis, 20 papers were selected, as shown in presented in Table 3.

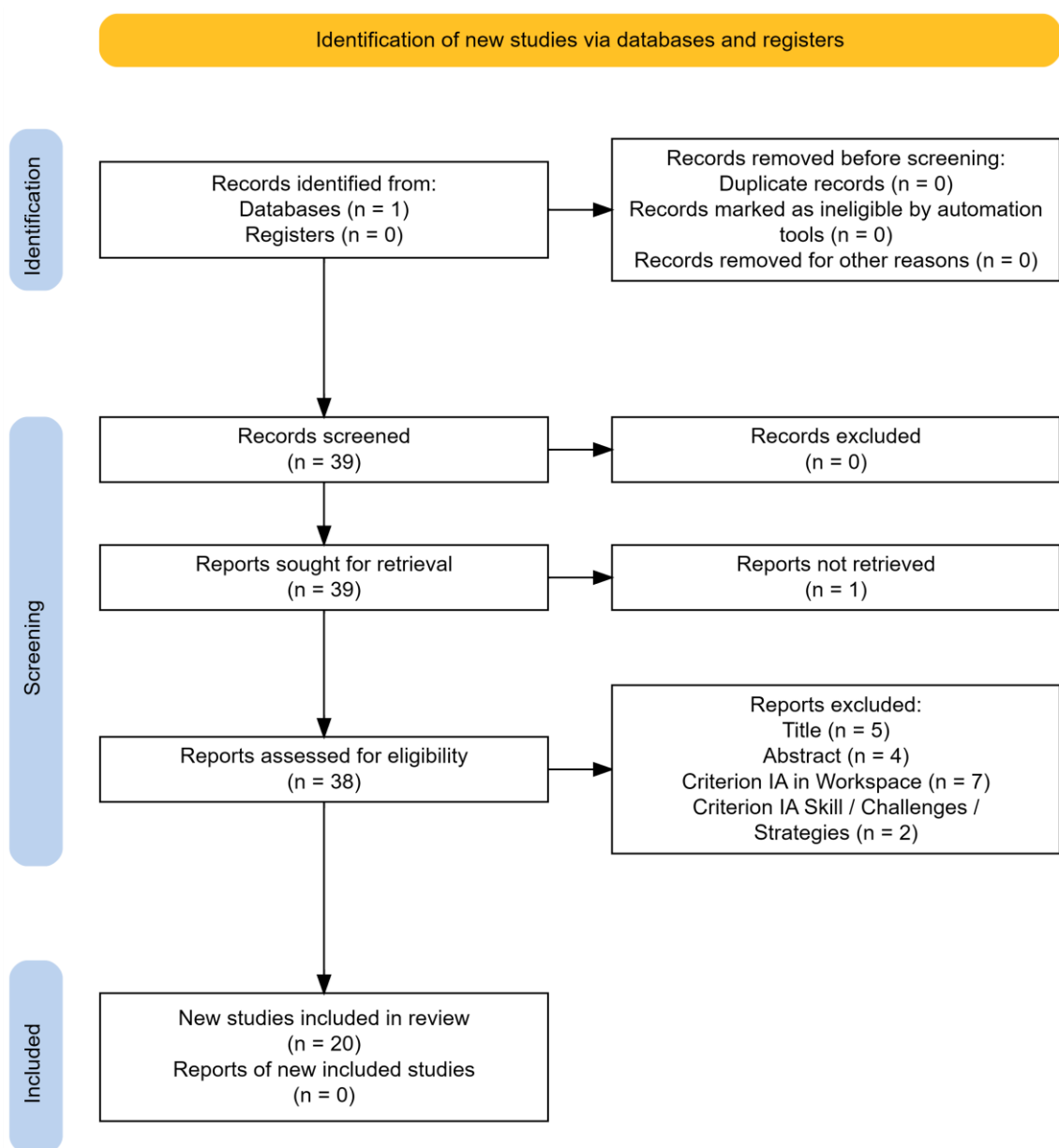


Figure 1. PRISMA flow diagram for this Rapid Review.

Table 3. List of selected papers.

Ref	Author	Year	Title
[1]	Kumar et al.	2023	The Current State of Software Engineering Employing Methods Derived from Artificial Intelligence and Outstanding Challenges
[2]	He et al.	2023	Linking employees' challenge-hindrane appraisals toward AI to service performance: the influences of job crafting, job insecurity and AI knowledge
[3]	Filippi et al.	2023	Automation technologies and their impact on employment: A review, synthesis and future research agenda
[4]	O'Donovan et al.	2023	Empowering future care workforces: scoping human capabilities to leverage assistive robotics
[5]	Bukartaite and Hooper	2023	Automation, artificial intelligence and future skills needs: an Irish perspective

[6]	Beebeejaun and Gunpath	2023	A Study of the Influence of Artificial Intelligence and Its Challenges: The Impact on Employees of the Legal Sector of Mauritius
[7]	Ibrahim and Al-Hiti	2023	Challenges of Employing Artificial Intelligence Technologies in Iraqi Media Institutions (A Field Study)
[8]	Faraj	2022	A Proposal to Employ Artificial Intelligence Applications in Developing Prince Sattam Bin Abdulaziz University Students' Future Skills
[9]	Lui and Shum	2022	Impact of Robotic Process Automation on Future Employment of Accounting Professionals
[10]	Baral et al.	2022	Role of Digital Technology and Artificial Intelligence for Monitoring Talent Strategies to Bridge the Skill Gap
[11]	Jurczuk and Florea	2022	Future-oriented digital skills for process design and automation
[12]	Chen et al.	2022	AI-Assisted Enhancement of Student Presentation Skills: Challenges and Opportunities
[13]	Aljohani et al.	2022	A Methodological Framework to Predict Future Market Needs for Sustainable Skills Management Using AI and Big Data Technologies
[14]	Farrow	2022	Determining the human to AI workforce ratio – Exploring future organizational scenarios and the implications for anticipatory workforce planning
[15]	Paško et al.	2022	Plan and Develop Advanced Knowledge and Skills for Future Industrial Employees in the Field of Artificial Intelligence, Internet of Things and Edge Computing
[16]	Jones et al.	2022	Automation in Colombia: assessing skills needed for the future of work
[17]	Anagnoste et al.	2021	The Role of Chatbots in End-To-End Intelligent Automation and Future Employment Dynamics
[18]	Chirgwin	2021	Skills development and training of future workers in mining automation control rooms
[19]	Saukkonen et al.	2021	Human Rights, Employee Rights and Copyrights – Parallels of AI Enablers and Obstacles Across Occupations in Human-Centric Domains
[20]	Li et al.	2020	Skill Learning Strategy Based on Dynamic Motion Primitives for Human–Robot Cooperative Manipulation

8. Summary of Articles

Kumar et al. [1] provide a comprehensive overview of the current state of software engineering, focusing specifically on integrating methods derived from AI. In their study, they explore the potential impact of AI on traditional roles within the field of software development. They explore concepts such as "software 2.0," where AI, particularly neural networks, may replace human-performed programming.

He et al. [2] investigate the impact of employees' assessments of AI within the service industry, focusing on the hospitality sector. Their study examines how these assessments influence job-related outcomes such as job crafting, job insecurity, and service performance. The researchers utilize a sample of participants with AI experience in the service sector and employ established scales to measure various constructs. The study also discusses their findings'

theoretical and practical implications, providing managers with recommendations and directions for future research.

Filippi et al. [3] examine the impact of automation technologies, such as AI and industrial robots, on employment. They conduct a review of existing literature, which highlights the intricate and uncertain nature of these effects. The authors find that the impact of automation varies across different levels of analysis, industries, and worker characteristics. Moreover, they discuss the potential implications of these findings for both management and public policy.

O'Donovan et al. [4] focus on the intersection between healthcare professionals and assistive robotics to identify the essential human capabilities for effective collaboration. Through a year-long co-design project, the researchers engage stakeholders, including healthcare professionals, to identify the required key knowledge, skills, and capabilities. The stress of the study is to understand and promote responsible, safe, and ethical use of physically assistive robotics in dynamic care settings.

Bukartaite and Hooper [5] navigate the impact of the Fourth Industrial Revolution on the future of work in Ireland, focusing on automation, AI, and the changing skills required for knowledge workers. The authors draw on insights from interviews with professionals to discuss the shift towards lifelong learning, changes in skill requirements, and the effect of automation on Irish Small and Medium-sized Enterprises (SMEs). The research aims to provide valuable perspectives for understanding and preparing for the evolving workforce in the age of automation and AI.

Beebeejaun and Gunpath [6] investigate the impact of integrating AI into Mauritius's legal sector on employees' performance and adaptability. The authors analyze the challenges, opportunities, and ethical considerations associated with using AI, drawing on interviews conducted with employees from 13 law firms. The primary objective is to clarify the perspectives of legal professionals regarding how AI influences their understanding, utilization, hurdles, and recommendations.

Ibrahim and Al-Hiti [7] examine the challenges encountered by Iraqi media institutions when adopting AI technologies. The authors explore the transformative impact of AI on information gathering, content creation, and audience engagement. Additionally, insights are gathered from media professionals and decision-makers in Iraq to comprehend their perspectives on the obstacles and the importance of incorporating AI technologies in the media landscape.

Faraj [8] proposes the integration of AI applications within the education system of Prince Sattam Bin Abdulaziz University (PSAU) to develop students' future skills. The author emphasizes the importance of AI in achieving sustainable development goals and evaluates the readiness of PSAU students for AI adoption. The study assesses the requirements for employing AI applications, explores the future skills necessary for the labor market, and outlines a proposal for aligning education with industry needs.

Lui and Shum [9] explore the effects of Robotic Process Automation on the future employment of accounting professionals, especially considering the accelerated adoption of RPA during the COVID-19 pandemic. The study investigates students' perceptions of RPA's influence on accounting careers and examines how education shapes these perceptions.

Baral et al. [10] discuss how professional services companies use digital technology and AI to address skill gaps. The study emphasizes the adoption of AI for large-scale skill development and forecasts a significant global expenditure in this area. The research methodology involves experiments, data analysis, and the testing of hypotheses. Both primary and secondary data collection approaches are employed to study skill gaps across various organizations and examine various models for skill gap analysis.

Jurczuk and Florea [11] shed light on the challenges faced by industries in the era of Industry 4.0 and smart factories, particularly highlighting the shortage of digital skills. The authors stress the need for an extensive digital skills framework adapted to the specific demands of the Factory of the Future. To address this issue, they employ a multi-method approach that includes a literature review, a survey-based competency framework, and an exploration of the evolving interaction between humans and machines. The research contributes to ongoing discussions on competency needs in the labor market and provides valuable insights for future research and educational pathways.

Chen et al. [12] investigate the use of AI to enhance university students' presentation skills, especially for those learning English as an additional language. The authors address the challenges associated with traditional teaching methods and propose an AI-assisted online platform. The study employs a mixed research approach, combining quantitative and qualitative methods, including a beta test with student participants from engineering and humanities disciplines, to evaluate the platform's effectiveness.

Aljohani et al. [13] examines how to align young Saudi professionals' skills with the job market, aiming to bridge the gap between required and acquired skills. The authors utilize big data and AI to predict future skill needs, particularly emphasizing the Vision 2030 goals. A case study analyzing the Saudi job market is presented, offering recommendations in data science, NoSQL, and software engineering. This research aims to assist students, universities, and employers in matching skills to job opportunities, focusing on the technical aspects. The study is part of the "Thriving Economy Rewarding Opportunities" project, which utilizes smart systems for knowledge sharing, skill mapping, and learning monitoring. The paper contributes to Saudi Arabia's education and training investment efforts to prepare youth for the digital future and the Fourth Industrial Revolution.

Farrow [14] explores the challenges and opportunities organizations face as they adapt to AI, considering different ratios of human-to-AI interaction in potential future scenarios. Employing a participatory approach, the study utilizes the "Futures Wheel" methodology in workshops involving diverse participants. The authors present five potential organizational scenarios, ranging from human-centric to AI-centric, and examine the implications of each scenario. The aim of this research is to guide organizations in anticipating workforce strategies and comprehending the consequences associated with decisions regarding AI implementation.

Paško et al. [15] present the educational landscape in the areas of AI, Internet of Things (IoT), and Edge Computing (EC) within European universities, with a particular emphasis on the Industry 4.0 era. The authors highlight the increasing significance of these technologies, identify gaps between academic offerings and industry demands, and explore the knowledge and perceptions of students. This research involves an extensive curricula review, developing a survey to evaluate student knowledge, and the subsequent analysis of survey results. The study's findings have implications for educational institutions, businesses, and policymakers,

providing guidance in shaping future curricula and addressing the evolving needs of the job market.

Jones et al. [16] assess the consequences of automation on the future of work in Colombia, specifically focusing on technological advancements such as AI and robotics. The authors adopt a distinctive approach to identify skill sets resilient to automation, emphasizing work activities centered around human interaction. By utilizing the *Clasificación Nacional de Ocupaciones* (CNO) dataset, the study combines expert judgment with machine learning to forecast the likelihood of automation for various occupations in Colombia. This research underscores the significance of incorporating Latin American viewpoints into the global discourse on automation.

Anagnoste et al. [17] address the role of chatbots in end-to-end intelligent automation within the context of rapid technological advancements and their influence on employment dynamics. Their objective is to guide organizations in making informed decisions regarding digital transformation, emphasizing factors that affect the successful adoption of automation solutions. The study extensively examines robotic process automation, AI, and the integration of chatbots for comprehensive automation. Additionally, a case study is included to illustrate the successful deployment of a chatbot in the healthcare industry.

Chirgwin [18] presents the skills development and training of workers in mining automation control rooms, specifically focusing on the role of Mine controllers. They highlight the industry's shift towards automation and the challenges posed by the lack of well-defined roles in this context. The research methodology includes qualitative data collection methods such as observations, interviews, content analysis, and audits. The study contributes to the field by addressing the challenges and opportunities in the skill development of mine controllers within the context of mining automation.

Saukkonen et al. [19] integrates three studies investigating the adoption of AI in Human Resource Management (HRM), primary health care, and music composition. The authors identify shared and specific opportunities and barriers in these domains by analyzing qualitative data from in-depth interviews with AI developers and experts. The study contributes valuable insights into the nuanced impact of AI adoption across different occupational fields, advocating for tailored approaches to maximize benefits and address challenges.

Li et al. [20] outline a skill acquisition strategy for collaborative manipulation between humans and robots, specifically focusing on exoskeleton robots used in rehabilitation training. The authors propose a two-level approach incorporating Dynamic Motion Primitives, Gaussian Mixture Models, admittance control, and an adaptive neural network controller. The objective is to facilitate compliant and safe interactions between humans and robots. The suggested strategy involves learning from demonstrations, addressing the challenges associated with multiple demonstrations, and ensuring effective robot behavior during collaboration.

9. Findings

A synthesis of the articles' findings is provided, aiming to gather insights into the varied integration of AI across industries and workforce sectors.

RQ1: How are businesses integrating AI technologies into their workforce?

According to Kumar et al. [1], businesses in the software engineering field are integrating AI technologies into their workforce and workflows, employing methods derived from artificial

intelligence and emphasizing AI's role across different phases of the software development life cycle. It analyzes the evolution and impact of AI in software engineering, proposing that AI, including neural networks, could replace human-performed programming. Key areas demonstrating AI's most notable achievements and potential in software engineering include the systematic evaluation of data in neural networks, structured analysis of big data pools, and the automation of routine tasks through algorithmic processes. The benefits of AI in shortening the development cycle, reducing expenses, and increasing productivity in software engineering are highlighted.

Based on Anagnoste et al. [17], companies in different domains are employing automation on cloud, on-premise, or through a hybrid approach, shifting their workloads to software robots, thus reshaping the old operating models, reducing routine tasks, and creating jobs that are better suited for humans. Messaging platforms are becoming the dominant communication channel worldwide and, combined with rising customer demand for self-services, empower companies to establish internal and external communications with zero need for a new user interface. Modern chatbots are impressive facilitators of human-to-robot interaction, and their market size is expected to continually increase in the coming years. Chatbots use advanced technologies such as Natural Language Processing and intent recognition to surpass language ambiguity and understand complex phrases. They can easily integrate with popular messaging applications like Messenger, WhatsApp, Slack, and Viber, and also connect with RPA robots and other technologies. These bots can create visitor profiles and customize responses to suit individual needs, initiate conversations, have a *personality*, continuously improve, and seamlessly transfer a conversation to a human operator when needed.

In Faraj [8], the focus is on integrating AI technologies in education to prepare students for the future labor market. They proposed using AI applications to develop students' future skills at the university, addressing five key areas: the learning environment, faculty members, courses, students, and graduates. In pursuit of efficiency and cost savings, Lui and Shum [9] indicate companies are turning to Robotic Process Automation, a trend notably observed in accounting and finance departments where many tasks are routine and rule-based. Adopting RPA is most prominent in tax services, followed by advisory and assurance services among large accounting firms. Implementing AI in the mining industry is encountering challenges, Chirgwin [18] emphasizes the critical need to balance attention between humans and machines. This balance is essential to fully realize the promised benefits of autonomous mining technologies.

Nowadays, companies are actively investing in comprehensive skill development strategies to harness the benefits of AI, as Baral et al. [10] specifically explore professional services companies to see how they are adopting strategic planning by focusing on skills. The emphasis on skill-based planning reflects a proactive approach by organizations to align their workforce with the evolving landscape of digital technologies, particularly in the realm of AI.

In the healthcare sector, O'Donovan et al. [4] highlight that the industry is increasingly adopting AI technology, aided by comprehensive frameworks and tools provided to hospital and health system leaders. These tools enable healthcare providers to work with assistive robotics, which leads to better care outcomes, increased productivity and efficiency in care delivery, and faster availability of life-saving treatments. As per Bukartaite and Hooper [5], firms are actively integrating AI, particularly for automation and Industry 4.0. It's widely acknowledged that the automation of repetitive and simple tasks is widespread, focusing on enhancing overall efficiency. Chen et al. [12] explores the implementation of AI in university education using an AI-assisted presentation platform. The platform is accessible through web browsers and enables

students to improve their presentation skills independently. It comprises two modules: "Learning Units" which provide tips and "Course and Assignment" for submitting presentations for evaluation by AI and grading by the teacher. This approach demonstrates the effective integration of AI in university settings, providing students with customized learning experiences and valuable feedback on their presentation abilities.

According to Beebeejaun and Gunpath [6], the legal sector is gradually adopting AI technologies. AI shapes the legal profession by assisting practitioners in various tasks, such as identifying bias, offering initial consultation solutions, expanding information scope, and predicting legal case outcomes. A paradigm shift is highlighted towards adopting new technologies based on algorithmic developments in the sector. Ibrahim and Al-Hiti [7] explore the transformative impact of AI on media work, encompassing the integration of artificial intelligence in information collection, content production, and audience interaction.

In line with Paško et al. [15], organizations strategically integrate AI technologies into their workforce and operations by aligning with market-driven trends, optimizing collaboration between technologies (technological synergy), and leveraging government support and programs. This approach effectively addresses industry concerns regarding the shortage of skilled professionals. In the context of robotics, as discussed by Li et al. [20], the focus is on applying AI to enhance human-robot cooperative manipulation, with specific attention to exoskeleton robots. The idea is that industries can adopt AI through robotic systems that learn and replicate human skills, facilitating safe collaboration between humans and robots. Implementing a skill learning-based hierarchical control strategy is highlighted as a valuable framework for industries aiming to integrate AI technologies into their workforce, particularly in tasks involving human-robot cooperation.

RQ2: What are the skill sets required for AI adoption?

According to Kumar et al. [1], software engineers must adapt to AI tools, continuously updating skills in language processing, machine learning, machine vision, big data, and IoT technologies. The emphasis is on ongoing skill development to stay current in the rapidly evolving field of AI. As Anagnoste et al. [17] indicate, combining technical skills, such as expertise in AI and chatbot development, with an understanding of business processes and workflows is essential for successful automation integration.

The proposal outlined by Faraj [8] emphasizes the importance of individuals acquiring essential technical skills, including proficiency in digital, scientific, technological, engineering, mathematical, and programming domains. Soft skills such as a commitment to lifelong learning and proficiency in digital culture are crucial for AI success. He et al. [2] underscore the significance of AI knowledge as a moderator influencing the consequences of AI evaluations on job insecurity and crafting. The use of rule-based RPA in Lui and Shum [9] requires accountants to develop new skills for collaboration with automated systems. This represents a shift in skill requirements towards working alongside technology and emphasizes the need for ongoing expertise development. For mine controllers, Chirgwin [18] in the context of AI adoption, exploring a mix of basic, social, and system skills is imperative. The importance of technical and emotional skills highlights a need for a combination of hard and soft skills.

Possessing the necessary skill sets is essential to integrate AI effectively [10]. Achieving this involves employing innovative learning methods, with a particular emphasis on experiential learning. The focus is bridging the skills gap in rapidly changing education and training sectors. This covers a range of expertise, including fundamental, technical, leadership, and digital

technology application skills. Based on recent findings by Filippi et al. [3], Workers must adapt to technological changes by acquiring new proficiencies, particularly digital skills, to minimize the risk of being replaced by automation. Higher education, technical skills, and proficiency in literacy and numeracy are key factors in reducing the risk of substitution.

O'Donovan et al. [4] identify six human capabilities healthcare professionals value for working alongside robotics and autonomous systems. These capabilities include physical and sensory skills (performing tasks and perceiving the environment), cognitive skills (thinking, reasoning, and problem-solving), social and emotional skills (Interacting with others and understanding emotions), adaptability (adjusting to new situations and learning from experience), systems thinking (understanding how different system parts work together), ethical decision-making (making ethical choices and understanding their implications). In the Factory of the Future context, Jurczuk and Florea [11] emphasized the critical importance of digital skills, encompassing areas like process design and automation. The competencies include technical and soft skills, such as creativity, critical thinking, digital content creation, data analysis, and cybersecurity.

Bukartaite and Hooper [5] focused on essential future skills, including programming, digital/ICT skills, data analytics, cybersecurity, creativity, communication, and emotional intelligence. The study underlines a shift towards soft skills, stressing the importance of balancing soft and hard skills, adaptability, and fostering a mindset that leads to innovation. In Chen et al. [12], the significance of oral presentation expertise is highlighted. Aljohani et al. [13] recommend skill sets for specific job sectors, encompassing machine learning, statistics, data visualization, web development, SQL development, IT management, and related proficiencies.

In the legal sector, Beebeejaun and Gunpath [6] underscore the need to provide law students with a foundational understanding of AI, integrating legal and technological knowledge. They suggest incorporating AI-related modules into curricula and offering continuous professional development courses. Jones et al. [16] identified skills that reduce the probability of automation, including communication, interpersonal relationship management, decision-making, and other soft skills. The emphasis on social expertise indicates that fostering soft skills could safeguard against the risk of automation. Improving technical skills among media workers is recommended by Ibrahim and Al-Hiti [7] to address the limited technical capabilities of AI. According to Paško et al. [15], future industrial employees should possess enough knowledge of AI, programming languages, and techniques including API, data processing, and big data management. The focus is on project-based learning, labs, and workshops to facilitate practical learning in AI, IoT, and Edge computing.

RQ3: What are the skill gaps or challenges they face in adopting AI?

In the context of skill gaps or challenges in adopting AI, as per Kumar et al. [1], AI's capabilities in automating routine tasks are constrained when solving unique problems and creating novel routines, necessitating human intervention. Faraj [8] identified obstacles in education encompassing infrastructure, faculty training, and program adjustments. They include weak trust between universities and industry, a shortage of qualified AI-focused faculty, reliance on traditional curricula, and financial constraints. He et al. [2] highlight employees' perception of AI as challenging and stressful, with job insecurity being a prominent concern. In the accounting field, Lui and Shum [9] explore differences in how students and experienced accounting managers perceive the impact of Robotic Process Automation on jobs. Entry-level accounting jobs may be replaced, requiring adaptation to technological changes in the industry.

Challenges faced by mine controllers, as outlined by Chirgwin [18], encompass a lack of clarity about their role, deficiencies in training and career advancement, concerns about mental and physical well-being, task overload, and gaps in organizational understanding. Baral et al. [10] indicate that Small and fast-growing firms face a "skills paradox" with frequent job changes, lack of education and training, and communication problems. Additionally, digital technology and AI encounter limitations due to networking issues, lack of training, and infrastructure challenges affecting AI effectiveness. Filippi et al. [3] point out that several factors impede the adoption of automation, such as labor costs, regulations, and societal preferences favoring human workers. Lower-skilled individuals exhibit resistance to automation, potentially resulting in a skills gap.

According to O'Donovan et al. [4], navigating robotics systems introduces challenges such as ambiguity in user requirements, uncertainties regarding autonomy, and the necessity for clear decision-making in system design and operation. Concerns about the suitability of technical features may also indicate potential gaps in understanding and utilizing robotic functionality. Jurczuk and Florea [11] highlight obstacles in digital transformation, encompassing a shortage of digital skills affecting the development of the Factory of the Future, demands for cognitive and domain-specific skills, and shifts in job requirements requiring new digital design competencies. These factors contribute to a gap in the education system. Skill gaps, particularly in critical thinking and self-management, create barriers to adopting advanced technologies. Bukartaite and Hooper [5] also note the challenge of predicting expertise due to the rapidly evolving nature of careers. A lag in technology integration, especially in SMEs, is attributed to mindset, financial resources, lack of awareness, and a mismatch between graduate skills and job market demands.

The education-related challenges identified by Chen et al. [12] encompass a grading burden for teachers, the demand for automated grading assistance, students' discomfort with oral presentations, particularly in engineering, and a call for additional training in delivery skills for teachers and students. Aljohani et al. [13] emphasize the lack of strategic research and planning for skill-building in the young generation and aligning young professionals' skills with the job market. Beebeejaun and Gunpath [6] outline obstacles to AI adoption within the legal sector, covering concerns about lawyers' autonomy, resistance to change, worries about data privacy, skepticism, and high acquisition costs. Additionally, challenges involve addressing cybersecurity risks and implementing safeguards against biased AI tools. As noted by Jones et al. [16], companies are grappling with the challenge of finding skilled human capital, underscoring the necessity for skills resistant to automation to mitigate potential skill gaps.

Hurdles encountered by Ibrahim and Al-Hiti [7] in media institutions involve weak technical capabilities, resistance to new roles, difficulties in verifying data sources, limited knowledge of AI capabilities, and difficulties in attracting talent. Organizational scenarios based on Farrow [14] pose challenges, including considerations of time context, implications of Human-to-AI ratio changes, potential biases, displacement of human workers, loss of natural decision-making and accountability issues, potential loss of human skills, potential social disruption, loss of empathy, and negative impacts on human well-being as well as concerns about AI becoming a "bully boss." Legal and ethical uncertainties in AI adoption are evident across various domains, as represented by Saukkonen et al. [19], specifically in HRM, primary health care, and music composition. Based on Paško et al. [15], there is a gap between industry needs and academic knowledge and a shortage of experts in IoT, AI, and EC applications. This includes low awareness of EC technologies, limited integration of cutting-edge technologies in

academic curricula, and knowledge gaps among students in applying AI, IoT, and EC to industrial problems.

RQ4: What are the strategies employed to address these challenges??

The first strategy in tackling the challenges of AI adoption, as proposed by Kumar et al. [1], emphasizes software engineers' continuous adaptation of skill sets. This adaptation is essential for effectively incorporating AI tools into their practices. In AI adoption, businesses must select solutions tailored to their current and future needs, ensuring a satisfactory return on investment. Anagnoste et al. [17] underscore the importance of aligning these solutions with employees' skills and workflows during the automation implementation process. This involves careful planning to integrate the automated workforce, manage its impact on human workers, and create pathways for new career opportunities. Ethical use of technology is crucial in this transformation.

In education, Faraj [8] outlines various strategies, including enhancing infrastructure, reconsidering academic programs, training faculty members and students in various future skills, creating partnerships with relevant institutions, and incorporating AI technologies into education. Practical approaches from He et al. [2] are recommended for adopting AI, including surveying employees' opinions on AI, encouraging a positive mindset towards obstacles, assisting employees in shaping their roles (job crafting), and customizing AI training for better performance. In addressing the challenges accountants face, Lui and Shum [9] suggest that accounting professionals should adapt to technological changes, emphasizing the importance of evolving accounting education to prepare students for the changing profession and the requirement for lifelong learning. Acknowledging the crucial role of human factors in mine control, Chirgwin [18] introduces a comprehensive Human Systems Integration (HSI) framework. This framework underscores a novel training approach considering hard and soft skills for mine controllers. Moreover, it sheds light on industry implications, underscoring the need for a stronger focus on skills development.

According to Baral et al. [10], businesses can employ various strategies, utilizing models such as Strengths, Weaknesses, Opportunities, and Challenges (SWOC) analysis, Gap Analysis Tools, McKinsey 7s, Nadler-Tushman's Congruence Model, and Burke-Litwin Causal Model. Additionally, Baral et al. [10] recognizes the critical role of digital technology and AI in effectively addressing the skill gap. Filippi et al. [3] suggest strategies, including restructuring work activities or providing training for workers. They underscore the significance of public policies encouraging firms to adopt automation technologies while ensuring worker protection. O'Donovan et al. [4] present methods for tackling obstacles such as system-wide training for various staff, legal teams, procurement officers, safeguarding and risk officers, and regulatory gatekeepers. Additionally, training and considering features of technology design, infrastructure configuration, and organizational culture are underlined. Jurczuk and Florea [11] indicates the requirement for a comprehensive digital skills framework tailored to the Factory of the Future's needs.

Bukartaite and Hooper [5] examine effective recruitment strategies, including prioritizing candidates with positive attitudes and learning agility, utilizing remote work to access a broader talent pool, and creating an appealing employee value proposition. Additionally, the authors advocate for talent sharing across companies, fostering partnerships, encouraging a culture of intrapreneurship, and promoting lifelong learning. These practices collectively enhance collaboration and productivity within the work environment. In developing an AI-assisted platform, Chen et al. [12] propose approaches that include designing an online AI-

assisted presentation training platform, using AI tools for assessment, conducting beta tests to examine reliability, and exploring student responses to identify areas of improvement. In Aljohani et al. [13], a significant emphasis is placed on the crucial role of digital technology and AI in addressing skill gaps.

In the legal industry, Beebeejaun and Gunputh [6] explore approaches that involve educating and training employees to embrace changes, establishing ethical guidelines for AI developers, and implementing public policies to promote the adoption of AI. Jones et al. [16] offer specific proposals for higher education, focusing on developing communication, social, and managerial expertise alongside technical expertise. Moreover, corporate leadership is advised to formulate strategies for skills development, and policymakers are encouraged to allocate resources to support workers at the risk of automation. In media institutions, Ibrahim and Al-Hiti [7] introduce methods that involve harnessing AI for information gathering, fostering collaborations with experienced entities, allocating financial resources, and coordinating educational initiatives and training programs.

Farrow [14] recommends analyzing macro themes through the Futures Wheel methodology to tackle challenges in AI adoption. This anticipatory approach to workforce design with a longer-term focus aligns workforce design with digital strategies, evaluating the ideal human-to-AI workforce ratio, addressing broader ethical concerns, developing models for anticipatory workforce design, investigating operational model changes needed for broader definitions of diversity, and supporting a scenario of *AI and Human Cooperation*. Regarding occupations in human-centric domains, there is an emphasis on customizing AI solutions to particular sectors rather than relying on generic approaches. Additionally, Saukkonen et al. [19] advocate for integrating ethical and legislative components into a revised Technology Acceptance Model.

According to Paško et al. [15], courses of action to address challenges involve developing or adjusting educational content, training a new generation of AI, IoT, and EC experts, aligning educational curricula with industry needs, forming partnerships with industries, supporting practical learning through internships, and incorporating real-world case studies in education. Projects like Planet4 contribute to bridging the gap between academic knowledge and industry requirements. In robotics, Li et al. [20] propose a two-tiered control strategy for human-robot teamwork. The upper level learns motor skills from human demonstrations using dynamic motion primitives and a Gaussian mixture model. At the lower level, the strategy ensures the robot's compliance during human interaction with admittance control and an adaptive neural controller. Experimental results demonstrate the strategy's success in cooperative tasks, showing effective learning and control for compliant robot-human interaction.

10. Conclusions

This rapid review highlights significant insights into the integration of Artificial Intelligence in the workplace and its implications for skill transformation. One of the key findings is the identification of a substantial skills gap that exists as businesses adopt AI technologies. The review reveals that while there is a growing demand for technical skills related to AI, such as machine learning and data analysis, there is an equally critical need for non-technical skills, including problem-solving, creativity, and ethical judgment. This dual requirement underscores the complexity of AI adoption, as it necessitates a multifaceted approach to workforce development.

Furthermore, this review discusses the various challenges organizations face in bridging this skills gap. Among these challenges, the rapid pace of technological advancement stands out, making it difficult for educational and training programs to keep up. Additionally, there is a notable lack of standardized curricula and certifications for AI-related skills, leading to inconsistencies in the skill levels of employees. The findings also highlight the cultural and organizational barriers that hinder effective AI integration, such as resistance to change and the need for a supportive leadership that understands AI's strategic importance.

In response to these challenges, several strategies that organizations are employing to facilitate the transition to an AI-augmented workforce were found. These strategies include investing in continuous learning and development programs, fostering collaborations between industry and academia, and adopting a more inclusive approach to AI training that goes beyond technical skills. By implementing these strategies, businesses can not only enhance their AI capabilities but also ensure that their workforce is equipped to thrive in an increasingly automated environment. Proactive and holistic approach to skill development is essential for maximizing the benefits of AI in the workplace, ultimately leading to improved productivity and innovation.

11. References

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